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December and January (Fig. 2). When temperature exceeded 13°C in early February, a coinciding increase in NO_3-N occurred in the midribs. Temperatures during the 1969 growing season were low in December, but were above the critical range during most of January and fell at or below the level for the remainder of the growing season. Again, the warming trend in January coincided with an increase in midrib NO_3-N. Further increases in midrib NO_3-N were obtained from subsequent fertilizer applications. These data point out that the effects of low temperatures on plant growth and N uptake cannot be overcome by applications of any N source used in this experiment until the temperature reaches or exceeds the threshold level. Based on the work of others (7) it is assumed that the differences in midrib NO_3-N levels due to N source could have been associated, in part, to the lack of mobility of the NH_4^+ions in the soil. Further, because higher NO_3-N concentrations were obtained with urea than $(NH_4)_2$-SO_4, some movement may have occurred before the urea hydrolyzed to NH_4^+ions. The slower nitrification rate with the sources containing NH_4-N was considered to be the cause for the initially lower NO_3-N level.

Evaluations for quality and disease factors showed no differences due to N source in either year.

Over 80% of the N in lettuce was taken up during the 4 weeks prior to harvest (Fig. 3). This is the period when the heads are forming and plant growth is rapid. The uptake curve is similar to that reported by Zink and Yamaguchi (8).

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Quantitative Chilling Enhancement and Negation in Peach Buds by High Temperatures in a Daily Cycle[1]

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Additional index words. rest, *Prunus persica*

Abstract. **The effect of exposure of small peach plants (*Prunus persica* (L.) Batsch cvs. Redhaven and Redskin) to daily fluctuating temperatures (16 hours low:8 hours high) of 6-15°, 6-18°, 6-21° and 6-24°C was examined in comparison with continuous chilling at 4° and a non-chilled control. Even though all chilling treatments received the same amount of chilling (weighted chilling hours), good leaf bud break was obtained only in the continuous 4°, 6-15° and 6-18° treatments. No lateral leaf buds opened in the 6-21°, 6-24° or the non-chilled controls of both cultivars. The 6-15° temperature regime was more efficient than continuous chilling in breaking bud rest on a weighted chilling hour basis. 'Redskin' lateral buds were shown to have a higher chilling requirement than 'Redhaven'. Terminal buds especially of 'Redhaven' showed a very low chilling requirement. Treating plants with 50 or 150 mg/liter GA_3 slightly advanced bud break but did not affect the level of sprouting.**

High temperatures are known to have a negative effect on rest development in dormant buds (1, 10, 14). The significance of high temperatures during rest was stressed by Weinberger (15) who pointed out that warm maximum temperatures during the main rest period controlled the rate of breaking rest in peach flower buds.

A marked negation of the chilling effect in peach vegetative buds was shown by Erez and Lavee (6) with temperatures of 21°C in a daily temperature cycle, but not when exposed to 18°. Richardson et al. (11) in their model for estimating rest completion, utilized a negative value for high temperatures. However, the coefficients they suggested are −0.5 chill units for temperatures of 16-18° and −1.0 for temperatures over 18°. The experimental basis for these values was not stated. The data of Erez and Lavee (6) suggest a stronger negative value for temperatures of 21°.

Under warm winter conditions, high temperatures frequently occur in addition to a lack of chilling temperatures. Therefore, both should be considered in developing a climatic model. The Utah model (11) fails to predict the end of rest for mild winter locations (Couvillon unpublished data) and there is a need for a more accurate quantitative determination of the high temperature effect on bud rest. The purpose of this work was to quantify the effect of chilling negation by high temperature in a diurnal cycle.

Materials and Methods

'Redhaven' and 'Redskin' shoots of current season's growth were removed from mature trees growing on the College Station farm in Athens, GA on July 31, 1977. The cuttings were treated as described earlier (4), then inserted in No. 3 vermiculite-filled

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10 x 10 x 6 cm plastic containers, and placed under intermittent mist. Root formation was evident within 2 weeks and the mist was gradually reduced during a 2 week period beginning from late August. Rooted plants (over 90% of the cuttings) were left in the greenhouse until November 15. Temperature extremes in the greenhouse were 30° maximum and 15° minimum. Leaf drop occurred by the end of October. On November 15, the plants were transferred to growth chambers for the treatments outlined in Table 1.

Table 1. Temperature treatments and periods of exposure.

Temperature conditions	Weighted chilling hours accumulated at time of sampling					
	Redhaven			Redskin		
24°C continuous	0	0	0	0	0	0
4° continuous	600	900	1200	500	750	1150
6-15° daily cycle[z]	440	900	1200	340	750	1150
6-18° daily cycle[z]	600	900	1200	500	750	1150
6-21° daily cycle[z]	600	900	1200	500	750	1150
6-24° daily cycle[z]	600	900	1200	500	750	1150

[z]16 hr at 6° and 8 hr at the high temperature.

Plants of both cultivars were held continuously at 24°C in the laboratory and at 4° in a cold room. Two bench-type and 2 walk-in type 'Sherer' growth chambers were used for the 4 cyclic temperature treatments. In all units high temperatures were maintained for 8 hr and low (6°) for 16 hr. Air temperature change from the high to the low levels and vice versa, required between 35 and 45 min. Bud temperatures, measured using copper-constantan thermocouples, lagged 5 to 7 min behind air temperatures following the temperature change. Variation from the designated temperature in each chamber was less than ± 0.4°. In the "6-15" chamber, 6° was not achieved during the first 14 days of exposure due to a technical failure. The low temperature reached for 16 hr daily was 9.5° which results in a difference in exposure time to chilling temperatures between "6-15" and the other treatments for the first sampling date (Table 1). The cumulative chilling value obtained for the 6-15° treatment was determined by calculating weighted chilling hours for the 14 day period as previously suggested (6). Root temperature required 2 to 2½ hr to equilibrate with the low air temperature level while at high temperatures a maximum of 1.0 to 1.5° lower than the chamber air temperature was obtained. During the 8 hr exposure to high temperature the plants were illuminated by cool white fluorescent light (4860 lux at plant level). The continuous 4 (±0.2°) treatment was maintained in darkness and the 24° (±1.5°) temperature

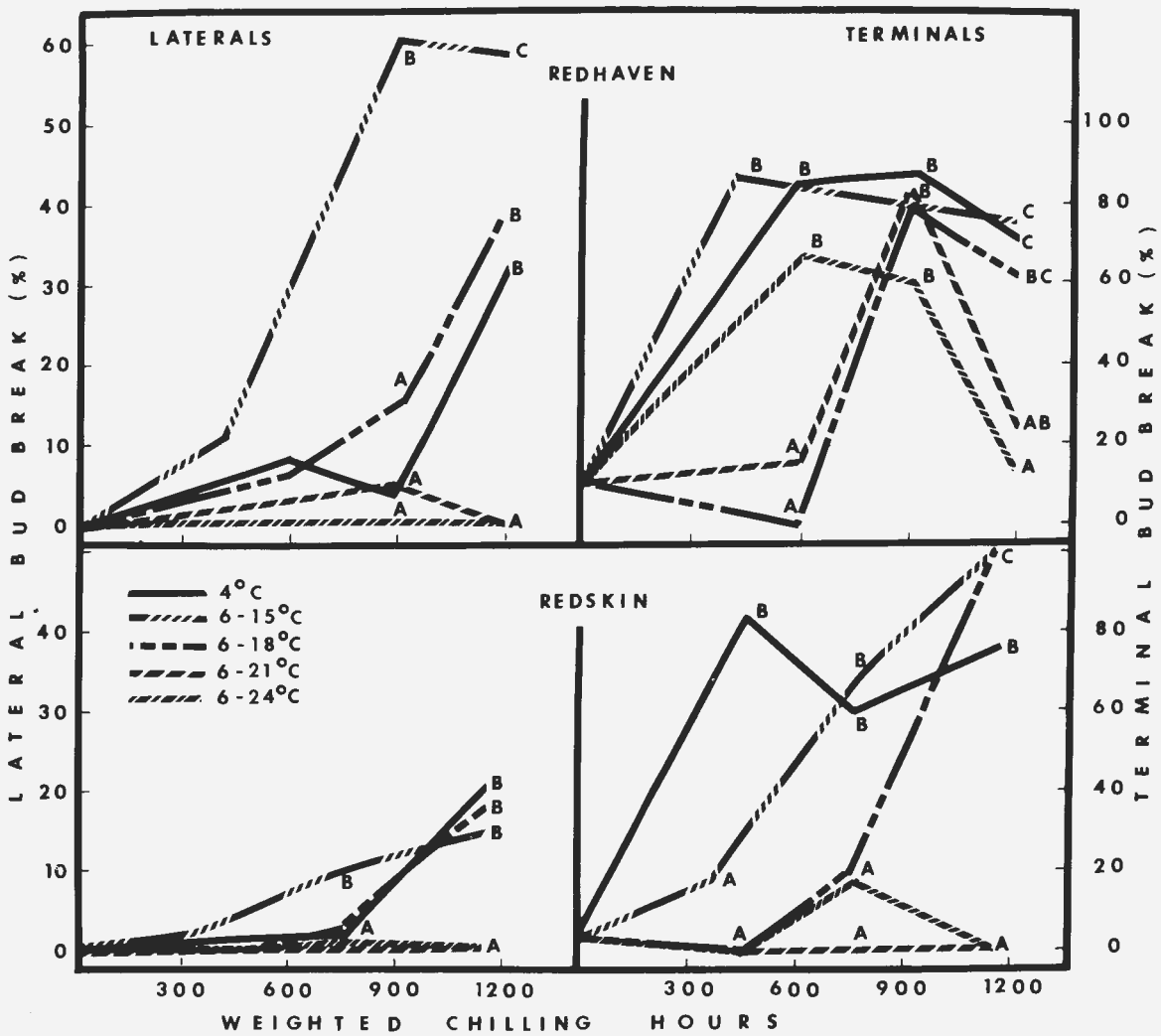


Fig. 1. Lateral and terminal bud break of 'Redhaven' and 'Redskin' peach following various chilling durations and temperature treatments. Means followed by a different letter at the same weighted chilling hour value differ significantly at P = 5% according to Duncan's multiple range test. The absence of a letter denotes an nonsignificant "F" value.

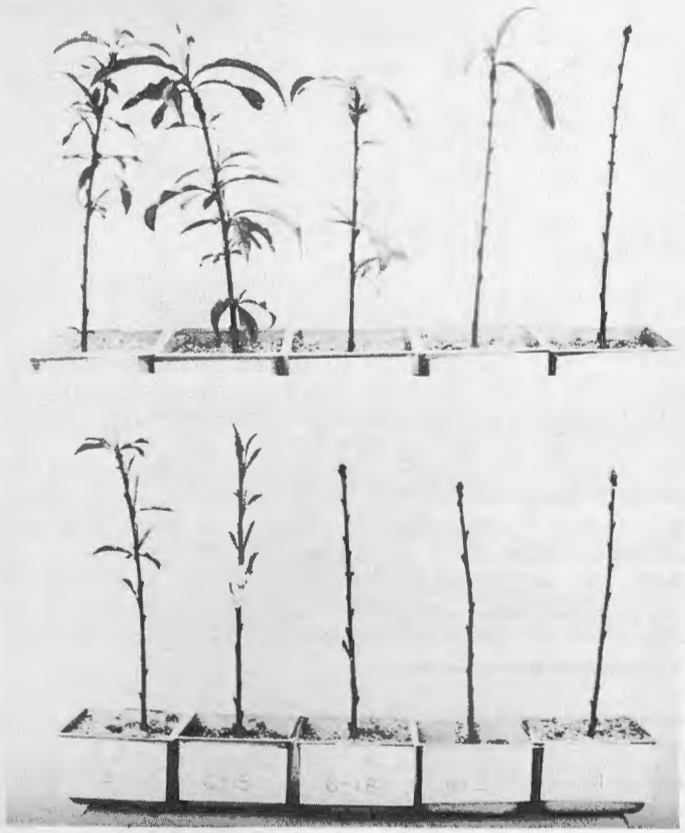


Fig. 2. Bud break on 'Redhaven' peach plants exposed to 1200 hr of chilling using various temperature treatments in comparison to a continuous 24° treatment after forcing at 23° for 14 (lower) and 30 (upper) days; 6-24° treatment was similar to the 6-21° treatment.

treatment was exposed to continuous white fluorescent light of 450 lux at plant level. Frequent watering eliminated drought stress to the plants throughout the experiment.

After completing their allocated chilling, the plants were transferred to 24°C under continuous light. On the last sampling date, plants transferred from each chilling treatment were divided into 3 groups that were treated with 0, 50, or 150 mg/liter of GA_3 containing 0.02% of a surfactant (X-77)[5] applied with a brush.

Every chilling duration and GA_3 concentration was applied to 6-8 individual plants. Lateral bud (LB) and terminal bud (TB) opening was recorded separately and results are presented as percent bud break after 14 and 30 days of forcing.

To determine the extent of high temperature antagonism to the applied chilling in the 6-21° and 6-24° treatments, plants of 'Redhaven' and 'Redskin' from all GA_3 treatments were moved following a 60 day period at 24° to continuous chilling at 4° for an additional 600 and 1200 hours. On transfer from continuous chilling back to forcing temperatures all TB were removed to eliminate the apical dominance effect.

Results

No lateral bud break occurred in the continuous 24°C treatment (data not presented).

Leaf buds of 'Redhaven' exhibited a better overall response to chilling than 'Redskin' indicating a lower chilling requirement (Fig. 1). 'Redhaven' showed high initial TB break for the 4°C, 6-15° and 6-24° treatments with 6-18° and 6-21° requiring 300 additional chilling hours to reach similar levels. In the 6-21° and 6-24° treatment TB break was substantially lower than the other treatments following the accumulation of 1200 chilling hours, indicating an increased inhibition induced by these conditions with duration of the treatment. 'Redskin' TB held at 4° and 6-15° opened best at 750 hr with 6-15° and 6-18° treatments showing similar levels at 1150 chilling hours. Poor TB opening occurred with the other treatments. However, since TB opening is based on 8 buds per treatment, the data are indicative only.

Substantial LB opening was obtained only at 6-15°, 6-18°, and the continuous 4° treatments in both cultivars (Fig. 1). A slower rate of bud opening was observed in the 6-18° than in the 6-15° or 4° treatments (Fig. 2). Low levels of bud break occurred with 'Redhaven' in 6-21° following the accumulation of 600 and 900 chilling hours. The 6-15° treatment was significantly more effective than either 6-18° or 4° treatments after exposure to 900 and 1200 chilling hours. With 'Redskin', a significantly higher level of bud break was obtained with 6-15° over the other treatments at only the 750 hour chilling accumulation level (Fig. 1).

The greater the GA_3 concentration the higher the rate of bud break obtained after 14 days of forcing in the 6-15° and 6-18° treatments (Fig. 3). Although significant differences were obtained with 'Redskin' following 14 forcing days, after an additional 16 forcing days these differences due to GA_3 treatment were lost (Fig. 3). GA_3 did not induce any bud break in the 6-21° or the 6-24° treatments for either cultivar.

In comparison to the continuous chilling treatment, similar levels of bud break were obtained after partial or full chilling levels were applied to the plants previously exposed to the 6-21° or 6-24° cycles (Table 2). These indicate a complete or an almost complete reversion of chilling by both 21° and 24° in a daily cycle.

Discussion

The effect of high temperatures, alternating with chilling temperatures in a daily cycle, on rest of LB peach buds was shown to be dependent on the level of the high temperature. While temperatures of 18° or lower did not antagonize chilling, temperatures of 21° and 24° resulted in full reversion of the chilling effect. This conclusion is also supported by the findings that GA_3 failed to induce bud break in the 6-21° or 6-24° treatments and that a low level of bud break was obtained upon exposure of partially rechilled plants from these treatments to forcing temperatures. This verifies the earlier results of Erez and Lavee (6) under conditions of greater temperature control and a wider temperature range. Also, these data emphasize the very drastic effect of high temperature on negating chilling. A threshold of high temperature negation of the chilling effect seems to exist between 18° and 21°, since a diurnal alteration of low temperature with temperatures above this threshold will completely negate the chilling effect while low temperature alternated with temperatures below this threshold had little or no effect on the negation of chilling.

Our data have a direct bearing on any climatological model simulating temperature effects on rest. The full reversion of the 6-21°C and 6-24° treatment shows that 8 hr of high temperature negated 16 hr of chilling. Thus, an hour at 21 to 24° and presumably higher in a diurnal cycle should have a value of at least −2, rather than the −1 coefficient suggested in the Utah model (11). Furthermore, since the negative effect of high temperature was obvious only above 18°, and in light of the fact that increased efficiency in releasing buds from rest was obtained with a diurnal cycle in which 15° was the maximum temperature, a value of −0.5 for temperatures of 15 to 18° range as suggested by Richardson et al. (11) seems incorrect.

Results obtained with the 6-15° treatment is of special interest regarding the interplay of temperature on resting peach buds. The alternation of chilling temperatures with 15°

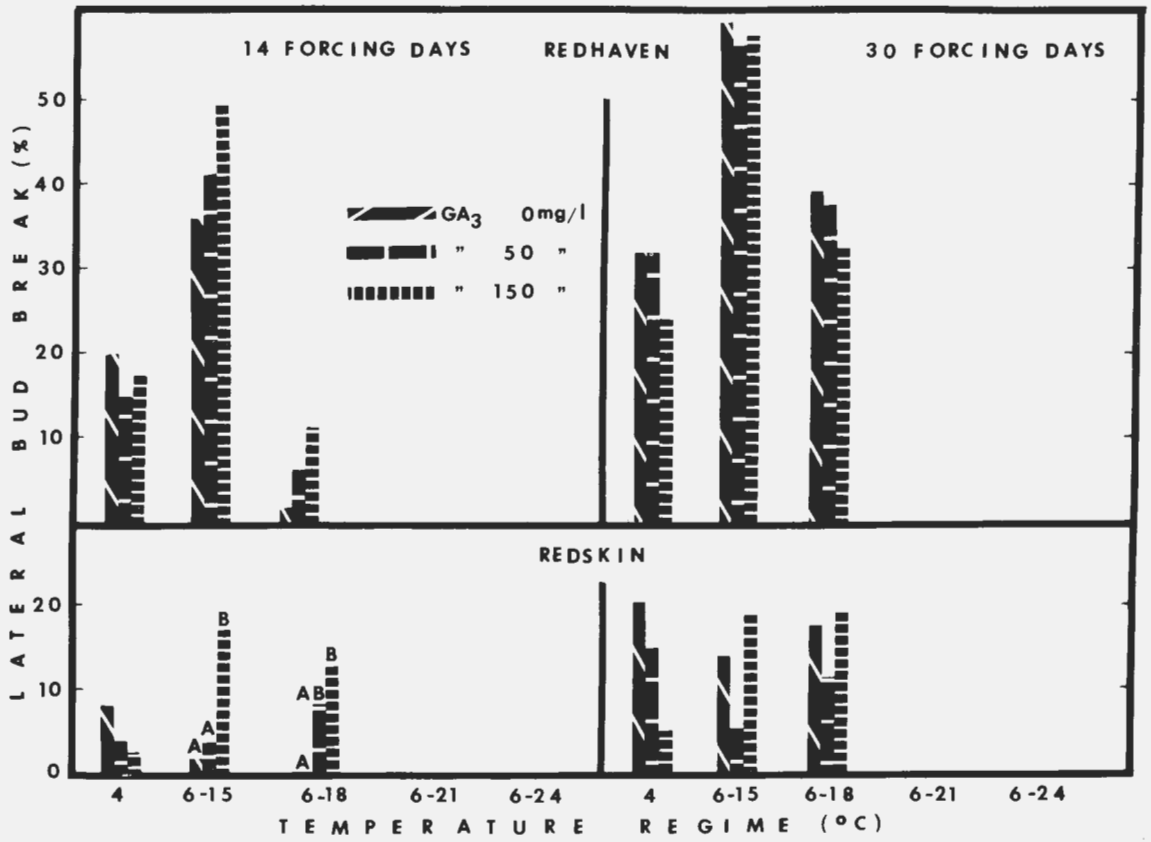


Fig. 3. The effect of GA_3 treatments on level of lateral bud opening in 5 temperature treatments. 'Redhaven' and 'Redskin' plants were exposed to 1200 and 1150 weighted chilling hours respectively. Means followed by a different letter at the same temperature treatment differ significantly at P = 5% according to Duncan's multiple range test. The absence of a letter denotes a nonsignificant "F" value.

in a daily cycle increased the efficiency of chilling with 'Redhaven' (the lower response of 'Redskin' at 6-15° could indicate a stage dependent response as 'Redskin' obviously has a greater chilling requirement which was not satisfied by any applied treatment). Since 15° by itself is not active in breaking rest (6) it would seem that the positive effect obtained resulted from its cycling with low temperatures.

The greater efficiency of alternating temperatures over continuous chilling reported here is supported in the literature. In a study by Bennett as quoted by Brown (2) a more rapid emergence from rest was obtained with pears when exposed to the "proper proportions of alternate chilling and growing temperatures than with continuous chilling." Also Brown (2, 3) reported a correlation between high minimum temperature in Davis, California and advancement of flower bud development in the apricot. He states that "the temperatures which are most effective in advancing bud development during the rest period are those which are low enough to break rest but still high enough to stimulate growth as the rest influence diminishes." An enhancing effect of moderate temperature on vernalization was reported by Friend and Gregory (8) in 'Petkus' winter rye. They relate it to an enhanced fixation effect. The relatively long chilling period required under continuous chilling (6) as opposed to field conditions (11) may be the result of a less efficient rest breaking effect of continuous chilling at low temperature. The consistently lower rate of leaf bud break obtained with 'Redskin', compared to 'Redhaven', indicates a higher chilling requirement for the former counter to that reported in the literature (12).

Table 2. The effect of preconditioning at 6-21° and 6-24°C daily cycle on the chilling requirement of the 'Redhaven' and 'Redskin' leaf buds. Bud opening determined after 30 forcing day.

Cultivar	Preconditioning		Hr. of chilling applied	Lateral bud break ± SE (%)
	Temp. regime	Chilling hours		
Redhaven	6°-21°C or 6°-24°	1200	600	5.3 ± 2.3
			1200	48.6 ± 5.9
	--	0	600	4.3 ± 3.0
	--		1200	32.1 ± 7.2
Redskin	6°-21° or 6°-24°	1150	600	0.5 ± 0.5
			1200	13.2 ± 3.2
	--	0	500	2.0 ± 2.0
	--		1150	20.1 ± 5.4

The effect of GA_3 on peach buds was to advance bud opening but not to increase the final level of bud break after 30 days of forcing. Similar data were reported earlier at comparable concentrations (7). Other reports in the literature on breaking rest by similar GA_3 concentration could not distinguish between advancing bud break and the actual effect on level of buds that broke due to the short "life" of the explants used as experimental material (5, 9). Also on whole plants, GA_3 was reported (13) to advance terminal bud break even though buds frequently force with no chilling.

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Flowering of Young Apple Trees following Summer Pruning[1]

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Additional index words. growth, fruit set, *Malus domestica*

Abstract **Axillary spurs and shoots produced in 1976 by heading back or pinching shoots did not flower in 1977 on 'Starkrimson Delicious' and 'Red Prince Delicious' apple (*Malus domestica* Borkh.) Similar treatments induced flower bud initiation on shoots of 'Cortland' when applied in 1976 and 1977 and on 'Red Prince Delicious' in 1977. Late June and early July treatments were more effective for inducing flower bud initiation than were treatments applied later. The year following summer pruning total bloom and fruit set were not increased and in some instances were reduced. Pruning current season shoots of 'Red Prince Delicious' and 'Cortland' to 0.5 to 1.0 cm stubs failed to induce flower bud initiation. 'Cortland' produced terminal flower buds on some long axillary shoots, while weak regrowth following pruning appeared necessary for formation of terminal flower buds on axillary growth of 'Red Prince Delicious'. Heading back most shoots on vigorous 'Cortland' and 'Red Prince Delicious' trees on June 21 or later devitalized the trees in spite of regrowth, although growth was not restricted the following year. Cultivar, tree vigor, and time and type of pruning all affected amount of regrowth.**

Summer pruning was practiced centuries ago by European gardeners to restrict vegetative growth and to induce the formation of flowering spurs (11, 16, 17), but has not been widely applied in commercial fruit growing. Response varied widely depending on time and type of pruning, environmental conditions, tree vigor, and cultivar (5, 6, 11, 13, 17). Results are difficult to interpret for lack of sufficient experimental description or statistical design, but summer pruning failed to suppress vegetative growth (11, 20), increase flowering (1, 14), induce early bearing (1), or increase production (1, 3) in some instances. In some of these studies, the procedure was similar to that practiced during the dormant season (1, 3, 11, 20), whereas European gardeners pinched shoots one or more times during the summer or broke shoots rather than removing whole branches or shoots (11, 13, 16, 17). Nevertheless, summer pruning generally restricted tree growth more than an equivalent amount of dormant season pruning (1, 5, 13).

Some early researchers concluded that results from summer pruning were too unpredictable and the practice too laborious to be of value in commercial American apple orchards (1, 5, 11, 13). Now, however, with closer spacing and the desire for quick return on investment, there is renewed interest in inducing flowering and controlling growth in young trees, especially in 'Delicious', which tends to make excessive growth and to be unfruitful (7, 8, 9). We have therefore re-examined the use of summer pruning to induce flowering in young apple trees.

Materials and Methods

1976. Trees used were vigorous, non-bearing 'Cortland'/Malling (M) 7A and 'Red Prince Delicious'/Malling Merton (MM) 106 planted in a commercial orchard at Wilbraham, Mass. in 1974 and 1975, respectively, and low to moderate vigor, light-bearing 'Starkrimson Delicious' (spur-type) on MM 106 established in a commercial orchard at Belchertown, Mass. in 1970. The pruning treatments on current season shoots were: (a) control-no treatment, (b) heading back to leave 4 to 6 leaves

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[3]Experiment Station.

and (c) pinching (removal of the succulent growing point). The shoots on every branch except the terminal shoot of the central leader were pruned. 'Red Prince Delicious' trees were pruned on July 1 or July 15, and 'Starkrimson' and 'Cortland' on July 1, July 15 or August 2. July 1 was 59 days after full bloom of 'Red Prince' and 'Cortland' and 64 days after that of 'Starkrimson'. A randomized block design was used with single-tree plots and 6 replications.

Shoot growth on 2 tagged limbs and trunk circumference were measured. In October, 10 to 13 of each of the following types of axillary growths below pruning cuts were marked on each tree: (a) buds that enlarged but failed to produce a whorl of leaves; (b) spurs with a whorl of leaves but less than 2 cm in length; and (c) shoots more than 2 cm in length. Bloom and fruit set in 1977 on the marked axillary growth and on 2 tagged limbs were determined.

1977. The 'Starkrimson' and 'Cortland' trees were not pruned in 1977 but growth on the tagged limbs and trunk circumference increase were measured, and bloom was determined in 1978. The 'Red Prince'/MM 106 trees were pruned for a second year; all current season shoots except the extension shoot of the central leader were headed or pinched on July 5 (61 days after full bloom) or July 19.

Vigorous, light-bearing 'Cortland'/M 7A trees in their 4th leaf in a commercial orchard at Wilbraham, Mass., and light-bearing 'Red Prince Delicious'/M 26 of moderate vigor in their 3rd leaf in a commercial orchard at North Brookfield. Mass. were summer pruned for the first time. Full bloom in Wilbraham was May 5, at North Brookfield May 10. Pruning of current season shoots on 'Cortland' was limited to 2 tagged limbs per tree. The shoots were pruned to stubs 0.5 to 1.0 cm in length, headed, or pinched on June 21, July 5 or July 19. On 'Red Prince'/M 26 all shoots with the exception of the terminal shoot of the central leader were headed or pinched on each pruning date and shoots were stubbed on July 5. Single-tree plots and 6 replications were arranged in a randomized block design.

Growth measurements were recorded as in 1976 except that buds merely enlarged were not tagged for identification in 1977, and trunk circumference of 'Cortland' was not measured.

Results and Discussion

Vegetative response. Researchers in the early 1900's reported that tree vigor influenced the amount of regrowth following summer pruning (10, 17). The correlation coefficient for the relationship between length of shoot at pruning and the length of regrowth was highly significant in 'Red Prince'/MM 106 trees pinched on July 5 ($r = 0.56$) or July 19 ($r = 0.59$) in 1977. Length of growth explained only about 32% of the variation in regrowth and therefore other factors not determined were of greater influence.

Shoots of 'Starkrimson' trees, a spur-type strain low to moderate in vigor, grew little. Therefore, little foliage was removed by pruning and growth was not restricted (Table 1). Heading removed much leaf surface in the vigorous 'Cortland' and 'Red Prince'/MM 106 trees. Despite regrowth (Tables 1, 2, 3), shoot growth per cm limb circumference and trunk circumference were reduced by heading, even when done in late June (Table 3). Heavy cropping of some 'Red Prince/M 26 trees in 1977 reduced vigor and caused variability in growth in response to pruning (Table 3).

Pinching generally did not restrict growth (Tables 1, 2, 3). Relative effects of pinching vs. heading on regrowth were inconsistent and in some instances the 2 methods produced comparable effects (Table 2, 3). Nevertheless, heading frequently forced more regrowth than pinching, whereas Gardner (10), using less severe pruning, observed that regrowth was associated with shoot growth at the time of purning, rather than severity of pruning.

Non-spur 'Delicious' strains are reported (9) to be more vegetative than non-spur strains of some other apple cultivars. The 'Red Prince'/MM 106 trees and 'Cortland' in this study were in adjacent rows, rather than being interplanted, but growth from 2 or more leaf axils following summer pruning was more common in 'Red Prince' than in 'Cortland' (Tables 1, 2, 3). The data for the 'Red Prince' M 26 trees which were less vigorous than the 'Cortland' support this observation (Table 3).

Stembridge (18) stubbed vigorous upright shoots in 'Delicious' trees in South Carolina to eliminate unwanted vigor, improve light penetration, and to induce flower initiation on regrowth below the pruning sites. Regrowth occurred from only 20% of the stubbed upright shoots pruned in the 'Red Prince'/M 26 trees on July 5, 1977 (data not shown). Less than 5% of the upright 'Cortland' shoots stubbed either on June 21, July 5, or July 19, 1977, produced regrowth. Regrowth occurred on stubs of vigorous vertical shoots, which are not common in Massachusetts until limbs of vigorous trees become horizontally oriented.

Vincent (20) in the early 1900's summer pruned when the terminal buds had formed to minimize regrowth; dates varied from August 6 to September 6 within a 7-year period. Pomologists currently suggest summer pruning in early August to restrict tree size and improve fruit quality (15, 19). Our primary objective was to re-examine the use of summer pruning to induce flower bud initiation on current season shoots rather than as a means of restricting vegetative growth. Nevertheless, the terminal shoots headed on July 1 were shorter than on the control trees (Table 4), and heading the majority of shoots restricted tree growth even when done as early as in late June (Table 3). Elfving (7) also found that heading cuts in early July reduced shoot growth. However, with non-spur trees of 'Delicious' and other highly vegetative cultivars (2), pruning in early summer may be undesirable because of proliferation of growing points behind the cuts on terminal shoots and vigorous upright shoots (Tables 2, 3).

Gardner et al. (11) suggested that the devitalizing effect of summer pruning might restrict growth the following year. This did not occur in our study (Tables 1, 2, 3). We found that 'Starkrimson' headed in 1976 made more shoot growth/cm limb circumference the following year than the control trees (Table 1) because growth per shoot was longer (data not shown).

Undesirable responses other than proliferation of growing points also occurred. On 'Cortland', 18% of the shoots headed July 15, 1976 were dead in 1977; death of headed shoots occurred less frequently as a result of pruning on July 1 or August 2. However, this adverse response was not encountered on 'Cortland' trees pruned in 1977. Death occurred in 50% of the 'Red Prince'/M 26 shoots stubbed on July 5, 1977, and 62 to 82% of the 'Cortland' shoots, depending upon date, similarly pruned on June 21, July 5 and July 19, 1977. The death of the headed shoots and that of the stubs was probably due to winter injury. Following pruning on August 2, 1976, some axillary 'Cortland' flower buds bloomed during September. Bloom also occurred in September on 1-year-old wood of 'Cortland' and 'McIntosh' following removal of current season shoots in early-August in a study at the Horticultural Research Center, Belchertown, Mass. (4). Lastly, leaf drop on control trees of 'Red Prince'/MM 106 on November 23, 1977 rated 4.4 (5 = complete defoliation) vs. 3.4 or less on headed or pinched trees (differences significant at 5%). Delayed leaf drop is an indicator of delayed maturity of wood, which increases the risk of winter injury and has been one of the concerns with summer pruning (7, 10).

Flowering and fruiting responses. Some spurs and shoots that developed directly behind the heading or pinching cut on